FINAL PROJECT REPORT

EV Market Performance & Insights

Tableau-Based Data Visualization and Descriptive Analysis

Field	Details
Project Title	EV Market Performance & Insights
Project Domain	Electric Vehicles / Data Visualization
Primary Tool	Tableau
Core Output	Interactive dashboard with 7 visualizations and a 3-point Tableau Story
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Submission Date	20-October-2026

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1. Introduction

1.1 Project Overview

Electric vehicles are an important part of the changing mobility landscape. EV datasets can contain a variety of information, such as vehicle categories, model years, manufacturers, battery technologies, driving ranges, customer ratings, and manufacturing countries. The EV Market Performance & Insights project uses Tableau to bring selected fields together in an interactive dashboard so users can explore and compare patterns in the available data.

1.2 Purpose

The purpose of the project is to make the available EV data easier to explore through clear visualizations, filters, and a short narrative sequence. The dashboard supports descriptive analysis and helps users inspect differences across the dimensions represented in the dataset.

1.3 Objectives

- Present seven relevant visualizations in one dashboard.
- Explore EV category distribution and annual sales trends by model year.
- Compare manufacturer representation and battery technology distribution.
- Compare average driving range and customer ratings across EV categories.
- Show the distribution of records by manufacturing country.
- Provide filters for selected dimensions and organize views into three Story points.
- Document the approach, checks, results, limitations, and future improvements.

1.4 Project Scope

The project focuses on Tableau-based descriptive exploration of the supplied dataset. It does not claim to provide live market monitoring, causal analysis, or predictive sales modelling unless those features are implemented separately.

1.5 Deliverables

- Tableau packaged workbook (.twbx).
- Dashboard containing seven visualizations and available filters.
- Tableau Story organized into three points.
- Business Question & Visualisation Report, Dashboard Design, and Story Design documents.
- Performance Testing document with actual evidence screenshots.
- Phase-wise project documentation and demonstration video.

2. Ideation Phase

2.1 Problem Statement

Users who need to explore EV market patterns may find it time-consuming to compare several metrics when information is spread across separate fields or views. A clear, interactive dashboard can help users compare categories, annual sales, manufacturers, battery technologies, driving range, customer ratings, and manufacturing-country representation in one place.

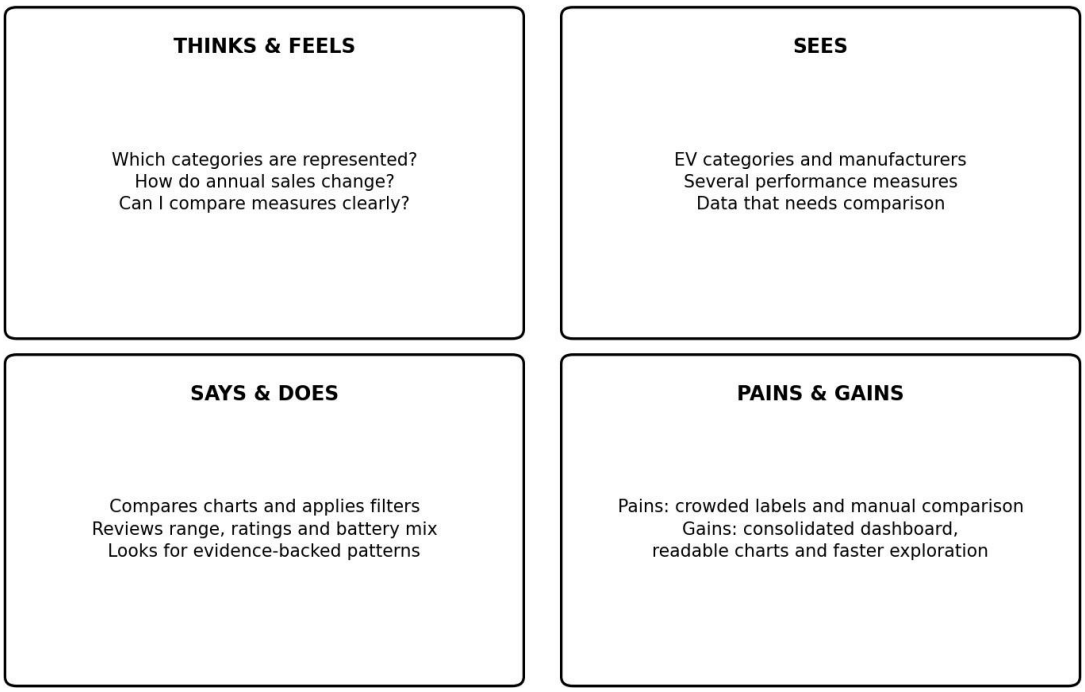
2.2 Target Users

- Data analysts exploring EV-related patterns.
- Business or market users comparing manufacturers and categories.
- Students or researchers reviewing the available EV dataset.

2.3 Empathy Map

The following is an illustrative planning persona based on the intended dashboard use; it is not presented as the result of formal interviews.

EMPATHY MAP — ILLUSTRATIVE EV MARKET ANALYSIS USER



2.4 Brainstorming and Idea Prioritization

Idea	Potential Benefit	Decision
Interactive Tableau dashboard	Consolidates key views and allows filter-based exploration.	Selected: fits current data and Tableau scope.
Static report with charts	Easy to share but offers limited interaction.	Useful as supporting documentation.
Predictive sales model	Could estimate future trends if suitable historical data and validation were available.	Not included in current scope.
Live market monitoring	Could reflect new data when a reliable source is connected.	Not included in current scope.

3. Requirement Analysis

3.1 Customer Journey

The expected user journey is to open the dashboard, understand the chart titles, compare the visualizations, use relevant filters, review the Story points, and communicate findings supported by the data.

3.2 Functional Requirements

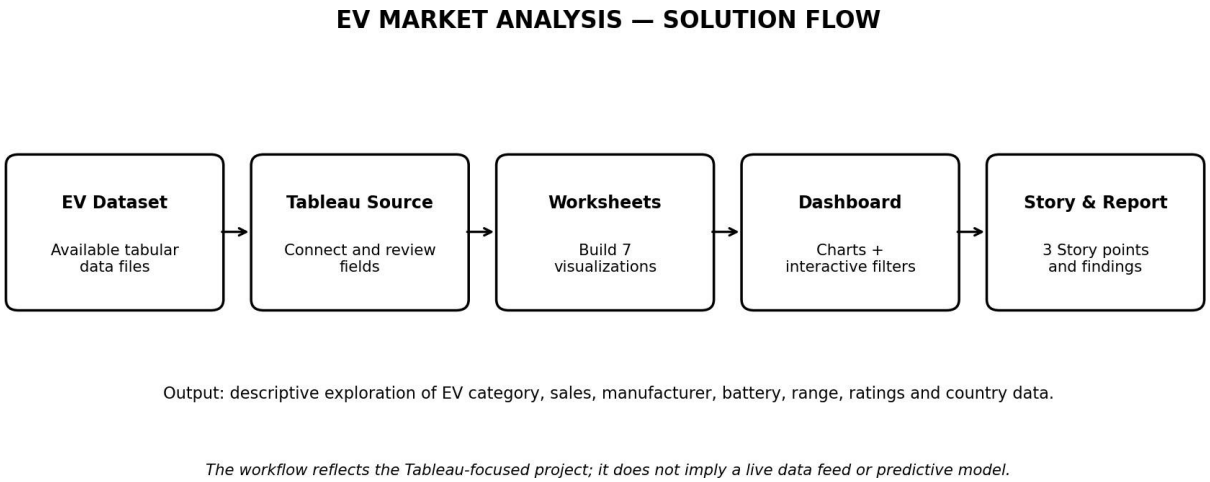
ID	Requirement	Verification
FR-01	Connect the available EV dataset to Tableau.	Confirm the active data source and fields in the workbook.
FR-02	Display seven EV-related visualizations.	All seven charts render and show the intended measures.
FR-03	Provide filters for relevant dimensions.	Test Category, Manufacturer, and Model Year filters as configured.

FR-04	Organize the analysis into three Story points.	Confirm each Story point opens correctly.
FR-05	Support reporting of descriptive findings.	Check claims against visible chart values/source data.

3.3 Non-Functional Requirements

- Readability: titles, labels, and axis values should be legible.
- Consistency: use clear names and consistent formatting.
- Accuracy: aggregations and values should match the connected dataset.
- Usability: filters should be visible and behave as expected.
- Maintainability: worksheets, dashboards, and Story points should have meaningful names.

3.4 Data Flow Diagram



3.5 Technology Stack

Layer	Tool / Asset	Purpose
Data	EV dataset files shown as ev_market_train and ev_market_test	Source data assets; confirm which file is connected to the final workbook.
Visualization	Tableau / packaged workbook (.twbx)	Build worksheets, dashboard, filters, and Story.
Documentation	Microsoft Word and PDF export	Prepare phase reports and final documentation.
Demonstration	Screen recording tool (to be confirmed)	Record the final walkthrough if required.

4. Project Design

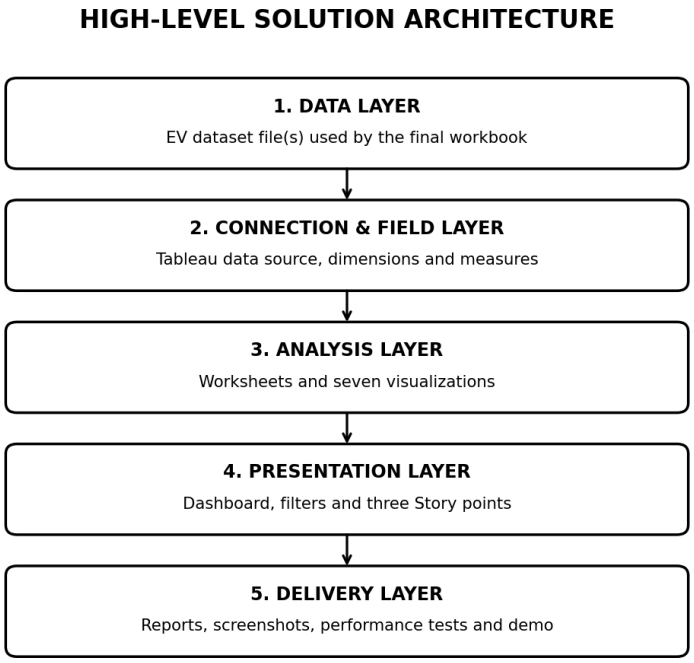
4.1 Problem Solution Fit

The solution aligns with the identified need by bringing several EV-related measures into a single interactive dashboard. The charts support comparisons across categories, model years, manufacturers, battery technologies, driving range, customer ratings, and countries. The solution is appropriate for descriptive analysis of the available dataset.

4.2 Proposed Solution

The proposed solution is a Tableau dashboard titled “EV Market Performance & Insights”, supported by a Story with three points: EV Market Overview, Vehicle & Battery Insights, and Manufacturer & Country Insights.

4.3 Solution Architecture



The architecture represents a high-level data-to-visualization workflow. It does not imply a separate database server, live API, cloud pipeline, or predictive model.

5. Project Planning & Scheduling

5.1 Main Tasks

Task	Expected Output	Status to Confirm
Confirm scope and questions	Business question list	Review final report.
Review dataset and fields	Confirmed source fields	Verify in Tableau.
Build seven charts	Seven visualizations	Visible in dashboard; final workbook check required.
Assemble dashboard and filters	Interactive dashboard	Test filter behavior.
Create Story sequence	Three Story points	Verify navigation.
Performance checks	Testing evidence	Replace placeholders with actual screenshots/results.
Documentation	Phase documents and final report	Review against official templates.
Demonstration and packaging	Demo video and final submission files	Complete before submission.

5.2 Suggested Schedule

The schedule below is an illustrative sequence, not a record of actual dates. Adjust it to the actual work completed and the submission deadline.

Suggested Period	Activity
Day 1	Confirm project scope and active data source.
Day 2	Review data fields and any required data checks.
Days 3–4	Finalize visualizations, dashboard layout, and filters.

Day 5	Verify the three Story points.
Day 6	Test chart rendering, labels, filters, and navigation.
Days 6–7	Finalize reports and evidence screenshots.
Day 8	Record the demo, export PDFs, and check submission folders.

6. Functional and Performance Testing

Testing should be performed on the saved Tableau workbook. The checks below describe the evidence to collect; mark a test as passed only after verifying it in Tableau.

Test ID	Test Area	Expected Result / Evidence
T-01	Data rendered	Data source is connected and the dashboard charts render.
T-02	Data fields	Relevant dimensions and measures are available; document actual cleaning only if performed.
T-03	Filters	Category, Manufacturer, and Model Year filters behave as configured.
T-04	Aggregations	SUM(Annual Sales Units), AVG(Driving Range), AVG(Customer Rating), and counts match worksheet configuration.
T-05	Dashboard	All seven visualizations are visible and readable.
T-06	Story	All three Story points open and show the intended views.

6.1 Screenshot Evidence to Insert

- Data Source tab or worksheet showing the connected dataset.
- Data preview or Data pane showing fields and data types.
- Full dashboard with filters visible.
- Worksheet shelves/Marks card showing aggregations, or calculation editor if custom fields exist.
- Full dashboard screenshot showing all seven visualizations.
- Story interface showing all three Story point tabs.

Testing limitation: this report does not assert that every test has passed. Replace this section's evidence list with actual screenshots and record pass/fail outcomes after testing.

7. Results

7.1 Project Output

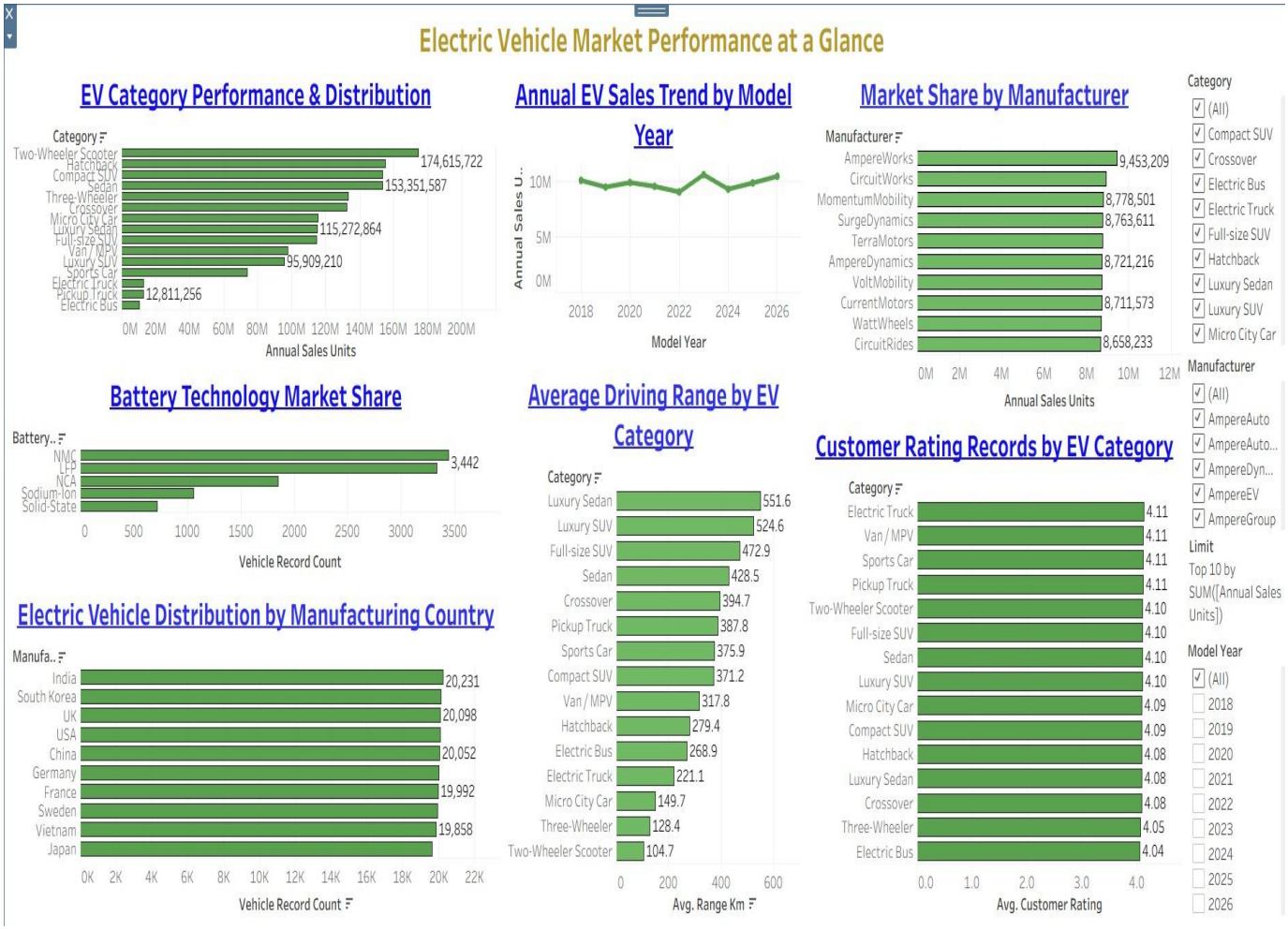
The project workbook contains a dashboard layout with seven intended visualizations and visible filters for Category, Manufacturer, and Model Year. The Story navigation shown during project work includes three points: EV Market Overview, Vehicle & Battery Insights, and Manufacturer & Country Insights. Confirm these elements in the final saved workbook before submission.

7.2 Visualizations Included

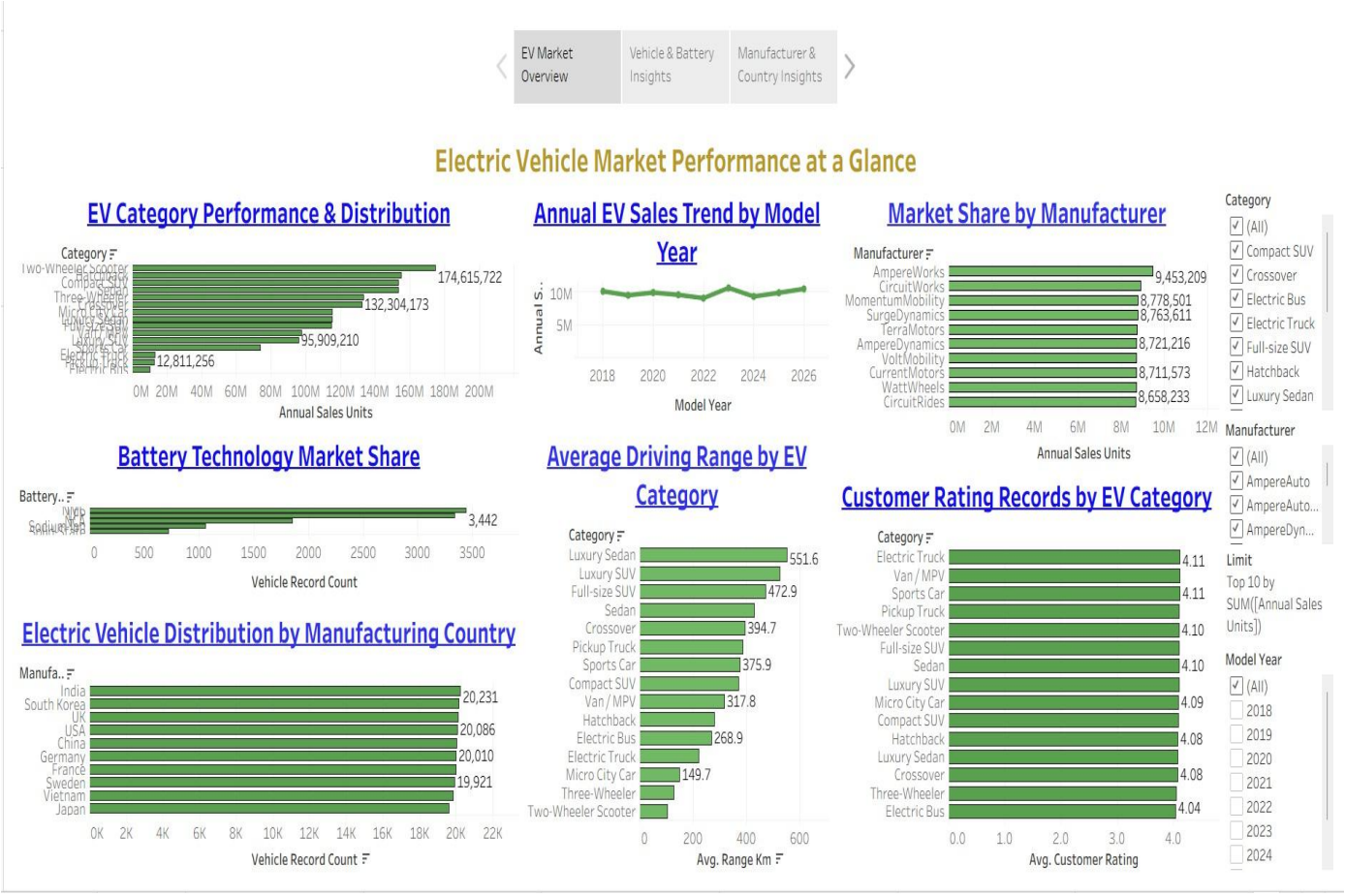
No.	Visualization	Purpose
1	EV Category Performance & Distribution	Compare the category-level distribution represented in the dataset.
2	Annual EV Sales Trend by Model Year	Inspect changes in annual sales across model years.
3	Market Share by Manufacturer	Compare manufacturers by annual sales units.
4	Battery Technology Market Share	Compare representation by battery technology.
5	Average Driving Range by EV Category	Compare average range across categories.
6	Customer Rating Records by EV Category	Compare average customer ratings across categories, according to worksheet configuration.
7	Electric Vehicle Distribution by Manufacturing Country	Compare records by manufacturing country.

7.3 Evidence Screenshots

- **Dashboard for Electric Vehicle Market Performance & Insights**



- **Story For Electric Vehicle Market Performance & Insights**



8. Advantages & Disadvantages

Advantages	Disadvantages / Limitations
Combines seven related views in one dashboard.	The quality of insights depends on the dataset's completeness and definitions.
Allows exploration through available filters.	Crowded labels can reduce readability and may require layout adjustments.
Uses visual comparisons that are easier to scan than raw tables.	Descriptive charts do not establish causation or predict future sales.
Story points organize views into a logical sequence.	No live updates are implied unless a refresh/source process is configured.
Can support project reporting and presentation.	Interpretation may be limited by the fields and records available.

9. Conclusion

EV Market Performance & Insights demonstrates how Tableau can be used to organize EV-related data into an interactive dashboard and a short Story. The seven visualizations focus on category distribution, annual sales, manufacturers, battery technology, driving range, customer ratings, and manufacturing-country representation. The solution supports descriptive exploration and communication of data-supported observations. Final claims should be validated against the saved workbook and underlying dataset.

10. Future Scope

- Improve chart label spacing, sorting, and readability based on final testing.
- Add clearly defined KPIs if the required fields and definitions are available.
- Add a date or time-based analysis if the dataset supports it.
- Connect a reliable updated data source if ongoing market tracking is required.
- Explore predictive modelling only if suitable historical data, a clear objective, and validation methods are available.
- Add accessibility improvements, definitions, and a short dashboard user guide.

11. Appendix

11.1 Project Files

File / Resource	Purpose / Note
EV vehicle market performance & insights.twbx	Tableau packaged workbook shown in the project folder; confirm which workbook is final.
Final EV vehicle market performance & insights project.twbx	Second packaged workbook shown in the folder; identify and submit the intended final version.
ev_market_train / ev_market_test	Dataset assets shown in the project folder; document the actual connected source.
Business Question and Visualisation Report	Phase 5 supporting report.
Dashboard Design Template	Dashboard layout and design documentation.
Story Design document	Story design documentation; verify its actual file extension.
Performance Testing	Performance testing document/PDF; add completed evidence screenshots.

11.2 Links

Dataset: [Ev_Market dataset].

GitHub repository: [https://github.com/samyakchoudhary919/Electric-Vehicle-Market-Research-Insights-project].

Demo video: [https://drive.google.com/file/d/1jwfbJXyp_gbScWU08iPQO-nR3KYXRIYU/view?usp=sharing].